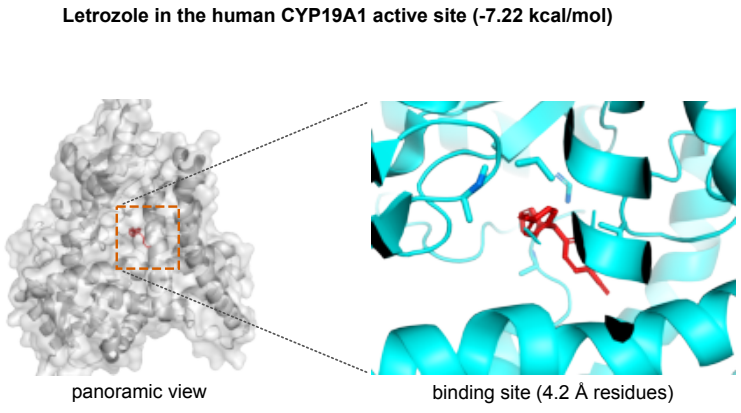
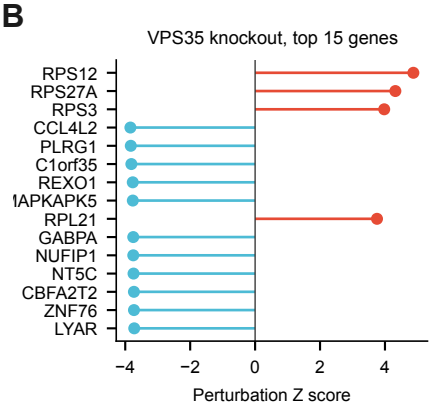
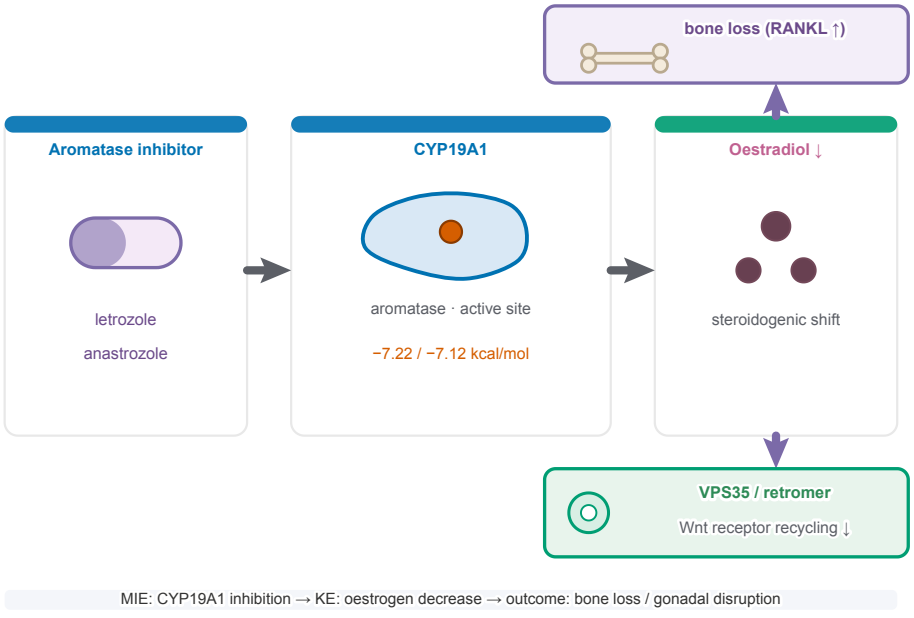
A


Letrozole in the human CYP19A1 active site (-7.22 kcal/mol)
panoramic view
binding site (4.2 Å residues)

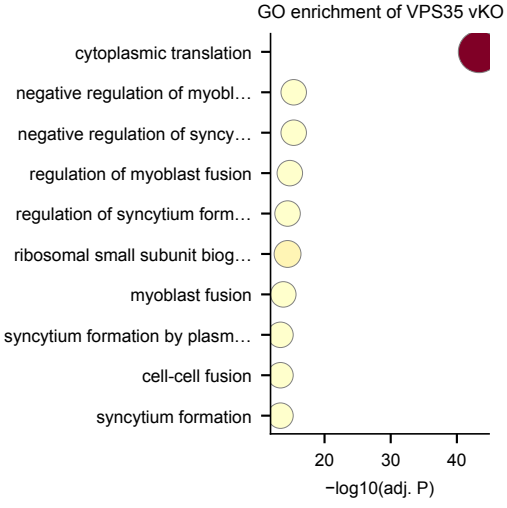
B


VPS35 knockout, top 15 genes
RPS12
RPS27A
RPS3
CCL4L2
PLRG1
C1orf35
REXO1
MAPKAPK5
RPL21
GABPA
NUFIP1
NT5C
CBFA2T2
ZNF76
LYAR
−4
−2
0
2
4
Perturbation Z score

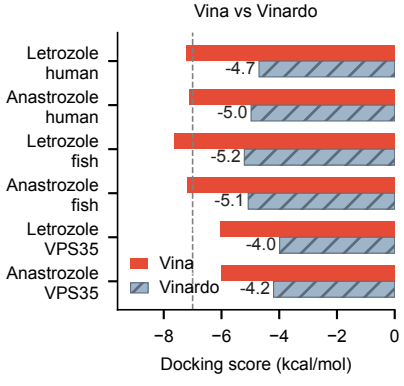
C


bone loss (RANKL ↑)
Aromatase inhibitor
letrozole
anastrozole
CYP19A1
aromatase · active site
−7.22 / −7.12 kcal/mol
Oestradiol ↓
steroidogenic shift
VPS35 / retromer
Wnt receptor recycling ↓
MIE: CYP19A1 inhibition → KE: oestrogen decrease → outcome: bone loss / gonadal disruption

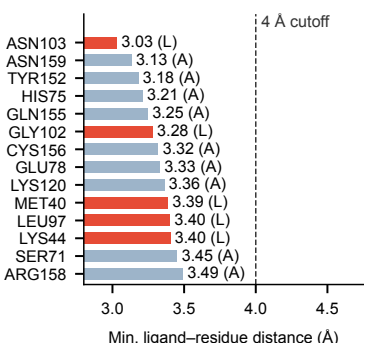
D


GO enrichment of VPS35 vKO
cytoplasmic translation
negative regulation of myobl…
negative regulation of syncy…
regulation of myoblast fusion
regulation of syncytium form…
ribosomal small subunit biog…
myoblast fusion
syncytium formation by plasm…
cell-cell fusion
syncytium formation
20
30
40
−log10(adj. P)

E


Vina vs Vinardo
Letrozole human
Anastrozole human
Letrozole fish
Anastrozole fish
Letrozole VPS35
Anastrozole VPS35
-4.7
-5.0
-5.2
-5.1
-4.0
-4.2
Vina
Vinardo
−8
−6
−4
−2
0
Docking score (kcal/mol)

F


VPS35 contact residues
4 Å cutoff
ASN103 3.03 (L)
ASN159 3.13 (A)
TYR152 3.18 (A)
HIS75 3.21 (A)
GLN155 3.25 (A)
GLY102 3.28 (L)
CYS156 3.32 (A)
GLU78 3.33 (A)
LYS120 3.36 (A)
MET40 3.39 (L)
LEU97 3.40 (L)
LYS44 3.40 (L)
SER71 3.45 (A)
ARG158 3.49 (A)
3.0
3.5
4.0
4.5
Min. ligand–residue distance (Å)